

Partial Differential Equations 3 (Numerical)

Lecture 9 - Academic Year 2025/2026

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Aims and Objectives

1. Scheme Design for Hyperbolic PDEs.
2. Numerical Difficulty in Solving Hyperbolic PDEs.
3. Total Variation Diminishing (TVD) Schemes.

Numerical Schemes for Hyperbolic PDEs

Recall that we consider a hyperbolic PDE of the form:

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0. \quad (1)$$

To allow us to consider different schemes easily, it is useful to set things up in a general way, rather than diving into any particular finite difference scheme. Hence, consider the Taylor expansion:

$$u(x, t + \Delta t) = u(x, t) + \Delta t \partial_t u(x, t) + \frac{\Delta t^2}{2!} \partial_{tt} u(x, t) + O(\Delta t^3). \quad (2)$$

Given the advection equation we consider, $\partial_t u(x, t) = -c \partial_x u$. Moreover, differentiating the advection equation with respect to t ,

$$\partial_{tt} u = \partial_t(\partial_t u) = \partial_t(-c \partial_x u) = -c \partial_x(\partial_t u) = -c \partial_x(-c \partial_x u) = c^2 \partial_{xx} u.$$

Substituting these relations to Eq. (2), we obtain

$$u(x, t + \Delta t) = u(x, t) - c \Delta t \partial_x u(x, t) + \frac{c^2 \Delta t^2}{2} \partial_{xx} u(x, t) + O(\Delta t^3). \quad (3)$$

At this point, it is useful to introduce an operator notation, and we will ‘hat’ the symbol/letter to mark that we ascribe it as a finite difference operator. In the case above (we ignore $O(\Delta t^3)$ terms), we can write the shorthand

$$\begin{aligned} u(x, t + \Delta t) &= \hat{L}_x u(x, t), \\ \hat{L}_x &= 1 - c \Delta t \hat{\delta}_x + \frac{c^2 \Delta t^2}{2} \hat{\delta}_{xx}. \end{aligned} \quad (4)$$

In this notation the scheme is first order accurate in time if we keep only the Δt term and second order accurate in time if we also include the Δt^2 term. The order of accuracy in space depends on what scheme we choose for the operator $\hat{\delta}_x$ and $\hat{\delta}_{xx}$. There are a number of possible schemes which differ in terms of the choices for $\hat{\delta}_x$ and $\hat{\delta}_{xx}$:

Name	$\hat{\delta}_x u$	$\hat{\delta}_{xx} u$	O(time)	O(space)
First Order Upwind (FOU)	$\frac{u_i^n - u_{i-1}^n}{\Delta x}$	0	1st	1st
Forward Time Centred Space (FTCS)	$\frac{u_{i+1}^n - u_{i-1}^n}{2\Delta x}$	0	1st	2nd
Lax-Wendroff	$\frac{u_{i+1}^n - u_{i-1}^n}{2\Delta x}$	$\frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2}$	2nd	2nd

In addition, there are other possible schemes that do not fit under the structure given in Eq. (4), such as

$$\begin{aligned}\text{Lax - Friedrich} : u_i^{n+1} &= \frac{u_{i+1}^n + u_{i-1}^n}{2} - c\Delta t \frac{u_{i+1}^n - u_{i-1}^n}{2\Delta x}, \\ \text{Leap Frog} : u_i^{n+1} &= u_i^{n-1} - c\Delta t \frac{u_{i+1}^n - u_{i-1}^n}{\Delta x}.\end{aligned}$$

To decide which scheme is best, we must balance four factors:

- Accuracy: we can determine the formal accuracy of the scheme in time and space from our derivation.
- Speed and efficiency: how does it perform?
- Stability: what choice do we have for Δt and Δx ?
- Ease of implementation: which scheme is easier to code?

Exercise 1: What is the order of accuracy in time and space for the Lax-Friedrich and Leap Frog methods?

Numerical Difficulty: Steep gradient or discontinuity

A well-known problem in solving hyperbolic PDE is how to deal with function with steep gradient or discontinuity. Let us illustrate this with an example. Consider an advection equation given by

$$\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} = 0, \quad \text{with } 0 \leq x \leq 1. \quad (5)$$

The initial conditions are:

$$u(x, 0) = \begin{cases} \sin^4 2\pi x & 0 \leq x < \frac{1}{2}, \\ 1 & \frac{4}{6} \leq x < \frac{5}{6}, \\ 0, & \text{otherwise,} \end{cases} \quad (6)$$

and we use periodic boundary conditions such that $u(x=0) = u(x=1)$. Note the discontinuities at $x = \frac{4}{6}$ and $x = \frac{5}{6}$.

Fig. 1 show example solutions of this problem using FOU and Lax-Wendroff schemes. For the FOU scheme, as it is first order accurate, the solution is inaccurate as can be seen for the $\sin^4$ part of the solution. This loss of accuracy is called dissipation in technical terms. The discontinuity becomes smoothed out but there is no spurious oscillations. For the Lax-Wendroff scheme, which is a second order accurate scheme, the solution is more accurate for the $\sin^4$ part. However, the presence of discontinuities in the initial conditions lead to unwanted/spurious oscillations. Technically this is called dispersion. These spurious oscillations can cause positive-definite fields, such as mass, to turn negative; thus, leading to instabilities.

In his 1959 paper SK Godunov states that all (linear) methods of greater than first order accuracy will produce spurious oscillations in the vicinity of large gradients. Sixty years after this statement, the difficulty of reconciling the absence of spurious oscillations and high accuracy remains a challenge that is not fully resolved for realistic problems. Most of the numerical methods applied to hyperbolic conservation laws are of second order accuracy, and they include some sort of treatment to remove the spurious oscillations.

These oscillations are due to Gibbs phenomenon, as illustrated in the figure below. Here, a square wave is represented by various numbers of terms in a Fourier expansion. Even if 100 terms are retained in the expansion, small over- and under-shoots remain.

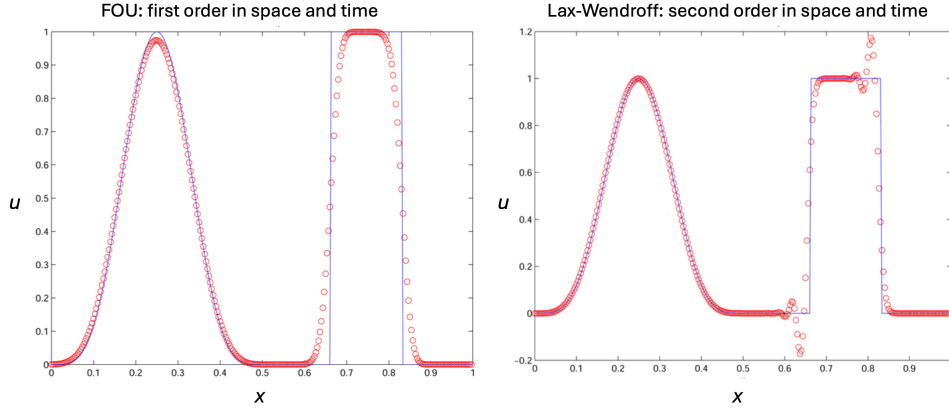


Figure 1: Numerical solutions of the example problem using FOU (First Order Upwind) and Lax-Wendroff schemes.

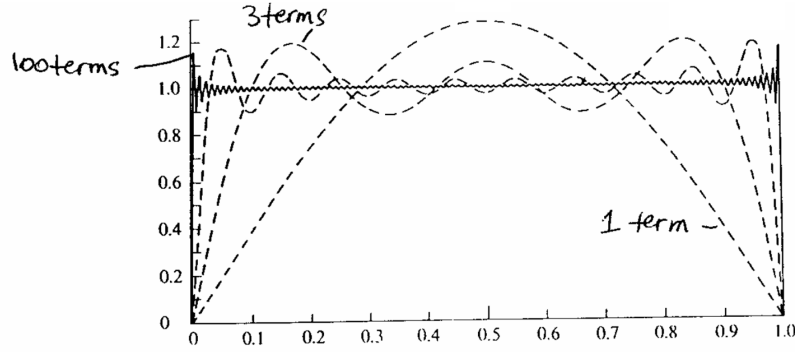


Figure 2: Illustration of Gibbs phenomenon, leading the difficulty to handle steep change/discontinuity in hyperbolic PDE.

Total Variation Diminishing (TVD) Schemes

In the mid 1980's Ami Harten published a paper which introduced the idea of total variation diminishing (TVD) schemes. These schemes take an oscillatory high order solution and apply a correction to the scheme which reduces the order of accuracy to first order in regions where oscillations would be generated, thus subverting Godunov's theorem. A key feature of TVD schemes is that they are monotonic, meaning no new extrema (minima or maxima) can be introduced into the solution as it evolves through time. This stops the spurious oscillations to develop. More formally, a useful quantity to consider is the total variation (TV) of the solution,

$$TV(t) = \int_{\Omega} |u(x, t)| dx. \quad (7)$$

A numerical scheme is said to be TVD if and only if

$$TV(t + \Delta t) \leq TV(t) \quad \text{for all } t, \Delta t. \quad (8)$$

Harten showed that if a numerical scheme could be written in the form

$$u_i^{n+1} = u_i^n - A(u_i^n - u_{i-1}^n) + B(u_{i+1}^n - u_i^n), \quad (9)$$

then a necessary and sufficient condition for the scheme to be TVD is

$$\begin{aligned} A &\geq 0, \\ B &\geq 0, \\ 0 &\leq A + B \leq 1. \end{aligned} \quad (10)$$

For example, we can show that the Lax-Wendroff scheme is not TVD. We can rewrite the

Lax-Wendroff scheme as follows:

$$\begin{aligned} u_i^{n+1} &= u_i^n - \frac{c\Delta t}{2\Delta x}(u_{i+1}^n - u_{i-1}^n) + \frac{c^2\Delta t^2}{2\Delta x^2}(u_{i+1}^n - 2u_i^n + u_{i-1}^n) \\ u_i^{n+1} &= u_i^n - \left[\frac{c\Delta t}{2\Delta x} + \frac{c^2\Delta t^2}{2\Delta x^2} \right] (u_i^n - u_{i-1}^n) + \left[\frac{c^2\Delta t^2}{2\Delta x^2} - \frac{c\Delta t}{2\Delta x} \right] (u_{i+1}^n - u_i^n) \end{aligned} \quad (11)$$

We can thus identify $A = \frac{c\Delta t}{2\Delta x} + \frac{c^2\Delta t^2}{2\Delta x^2}$ and $B = \frac{c^2\Delta t^2}{2\Delta x^2} - \frac{c\Delta t}{2\Delta x}$.

Using the $0 \leq A + B \leq 1$ set out by Harten, we get that $\frac{c^2\Delta t^2}{\Delta x^2} \leq 1$ or $\frac{c\Delta t}{\Delta x} \leq 1$.

At the same time, using the condition $B \geq 0$, we get that $\frac{c\Delta t}{2\Delta x}(\frac{c\Delta t}{\Delta x} - 1) \geq 0$, which leads to $\frac{c\Delta t}{\Delta x} \geq 1$. We find that the two conditions conflict and hence the Lax-Wendroff scheme is not TVD.

To date, there are a number of TVD schemes that have been proposed in the literature. Some of them are completely new schemes developed with TVD criterion in mind. Others modify existing schemes like Lax-Wendroff to make them TVD.

Additional Materials:

- Interactive tool: <https://kusumaatmaja.github.io/PDE3/examples>
- SK Godunov (1959) A Difference Scheme for Numerical Computation of Discontinuous Solution of Hydrodynamic Equations, Math. Sbornik 47:271–306.
- A. Harten (1984) On a Class of High Resolution Total Variation Stable Finite Difference Schemes, SIAM Journal of Numerical Analysis 21(1):1–23.

Version Control:

- version 1.0 on 20 December 2025.